Literature Review Guide Questions – Answers

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Topic: Cybersecurity threats in IoT for the Power and Utilities sector (smart grids and SCADA)

1) Focus and aim, audience

• Focus: risks to security from using IoT and IoT devices in smart-grid and SCADA setups.
• Aim: list main threat types and weak spots; pull together ways to reduce risks through rules, measures, and designs; spot gaps in studies for power company staff and rule-makers.
• Audience: grad students in research, safety experts in power firms, and policy groups.

2) Need and significance

• Updating the grid adds many devices, wireless connections, and cloud storage; chances of problems rise, and so do their effects.
• Keeping key operations safe calls for rules tailored to the field, not just basic IT protection; facts must guide buying, setup, and checks.

3) Context, perspective, and synthesis framework

• Context: mix of operations tech and IT, more use of IP networks, and rules pushing for strong systems.
• Perspective: based on real work, blending safety design with limits in power operations.
• Synthesis framework: group by themes like (a) attacker aims, (b) key items or areas, (c) protection levels; link to safety rules and traits such as uptime, wholeness, and security.

4) Source location and selection

• Databases: IEEE Xplore, ACM DL, ScienceDirect, IET, and NIST/ENISA stores.
• Keywords: 'smart grid security', 'SCADA cyber security', 'IEC 62351', 'NIST SP 800-82', 'IoT utilities', 'AMI threats'.
• Criteria: from 2010 to 2025; checked studies, known rules, and reports from watchdogs; favour hands-on or site-based work; focus on power sector.

5) Structure of the review

• Reasons and range
• Way to find and check proof
• Setup of smart-grid/SCADA and trust lines
• List of threats and real examples
• Range of measures and rules
• Check of how well they work and costs
• Gaps and next steps
• Wrap-up and suggestions

6) Main findings

• Threats: breaches at device ends (sensors, RTUs), breaks into substation networks, misuse of protocols, malware from suppliers, fake time signals, and data theft.
• Impacts: loss of sight or command, faulty safety plans, privacy leaks in AMI, and chain failures.
• Measures: layers of defence with network splits, safe protocols (IEC 62351 set), list of assets, strict updates or fixes for old gear, key handling with codes, spot checks for odd behaviour, and practice runs for events.
• Proof: solid advice from rule groups; more checks in real settings, but still few on how well measures hold up.

7) Strengths and limitations of the literature

• Strengths: well-set rules and model designs; plenty of overviews; more examples from sites since 2015.
• Limitations: few long-term data from power firms; not many tests that can repeat on live operations gear; updates and life cycles get little study.

8) Discrepancies

• Different takes on full encryption from field devices.
• Varied results on how well intrusion detection works in tight operations networks.
• Arguments over moving cloud for quick control against keeping all on-site.

9) Conclusions and recommended actions

• Put first on seeing assets, safe splits, and protocols that match rules, plus ongoing checks fit to operations patterns.
• Fund plans to shift old devices, flexible code use, and test attacks with safety nets.
• Study needs: side-by-side looks at detection tools with machine learning in operations data, key handling for big AMI, and ways to measure system strength.

References

ENISA (2017) Baseline Security Recommendations for IoT. Athens: European Union Agency for Network and Information Security.

Fang, X., Misra, S., Xue, G., and Yang, D. (2012) 'Smart Grid — The New and Improved Power Grid: A Survey', IEEE Communications Surveys & Tutorials, 14(4), pp. 944–980.

IEC (2018) IEC 62351: Power systems management and associated information exchange — Data and communications security. Geneva: International Electrotechnical Commission.

Knowles, W., Prince, D., Hutchison, D., et al. (2015) 'A survey of cyber security management in industrial control systems', International Journal of Critical Infrastructure Protection, 9–10, pp. 52–80.

NIST (2014) NISTIR 7628 Revision 1: Guidelines for Smart Grid Cyber Security. Gaithersburg, MD: National Institute of Standards and Technology.

NIST (2015) Guide to Industrial Control Systems (ICS) Security (SP 800-82 Rev. 2). Gaithersburg, MD: National Institute of Standards and Technology.

University of Essex Online (2024) 'List of Topics for Literature Review'. Module resource.

University of Essex Online (2024) 'e-Portfolio Activity: Research Proposal Review'. Teaching material.

University of Essex Online (2025) 'Unit 3: Methodology and Research Methods'. Teaching material.

University of Essex Online (2025) 'Unit 3 Seminar: Peer Review Activity'. Seminar brief.